

TAN-7

SCARA Robot (Selective Compliance Assembly Robot Arm)

TIMELINE

1. **Week-1:**
 - a) Research on basic working principle and utility of SCARA Robot
 - b) Selection & Procurement of Materials
2. **Week-2:**
 - a) Finalization of Mechanical Design of SCARA
 - b) Deliberation on possible innovations & consensus on making a **color classifier**
3. **Week-3:**
 - a) Supporting rods reduced to appropriate length & diameter in Fabrication & Machine Shop
 - b) Gripper design chosen & part models given for 3D Printing
4. **Week-4:**
 - a) Arduino Case designed in Woodworking Shop
 - b) Stepper Motor & Servo Motor Testing in Digital Manufacturing Lab
5. **Week-5:**
 - a) Base Fabrication in Woodworking Shop & Sheet Metal Shop
 - b) Hole drilling in Fabrication Shop
 - c) Code finalization and debugging done on Arduino IDE (Digital Manufacturing Lab)
6. **Week-6:**
 - a) Mechanical Assembly using materials from Fabrication Shop
 - b) Electrical Connections in Digital Manufacturing Lab

MATERIAL SELECTION

Reasoning behind using hybrid motor configuration:

NEMA-17 Stepper Motor used for providing the 4 degrees of freedom:

- Uniquely provide their maximum holding torque when stationary. This ensures the robotic arm remains perfectly rigid and locked in place.
- 200 steps per revolutions: utilize micro stepping to achieve high positional accuracy without the complex programming and wiring overhead of closed-loop encoders.
- Specializes in delivering high torque at low speeds.

Alternatives that were discarded:

- DC motors lack the positional accuracy.
- Servos lack the continuous rotation & rigidity required. Constant jittering would occur under the bulky mass.

Servo Motor used for gripper:

- Servo is incredibly lightweight, minimizing the payload stress on entire mechanism
- Gripper does not require continuous 360-degree rotation or step-counting. It only requires simple, absolute positional control.

Reasoning behind using lead screw for vertical motion:

We selected a lead screw for the vertical Z-axis instead of a belt-drive system because lead screw converts the rotational motion of the stepper motor into high-torque linear thrust, easily lifting the mass of the arm. Depending on the thread pitch, lead screws inherently provide a self-locking mechanism. This prevents the SCARA arm from back-driving or crashing down under its own weight when the stepper motors are unpowered.

Reasoning behind using stainless steel & aluminum for support structure:

Two of the four support pillars are made from stainless steel. Steel's high Young's modulus provides the necessary rigidity to handle the primary bending moments generated by the arms. These two rods act as the main structural backbone and the low-friction guide rails for the Z-axis linear motion. We substituted the other two support rods with aluminum to optimize for machinability without sacrificing the overall rigidity. We lacked more stainless-steel rods in the exact required diameter. Machining down a larger stock stainless steel rod over that length presents significant challenges regarding specific tools and surface finish degradation. Aluminum was selected for the other two rods because of its vastly superior machinability. This substitution solved a manufacturing bottleneck without compromising the structural integrity. Alternate rods of aluminum and steel were attached for balancing weight and structural stability.

Reasoning behind using different kinds of bearings:

To manage the complex, multi-directional forces generated by the SCARA mechanism, we utilized a combination of Thrust, Radial, and Linear bearings across the joints. By stacking a thrust bearing to support the weight and a radial bearing to support the bending moment, we ensured the motors only have to overcome the rotational inertia of the arm, rather than fighting friction.

- A) Thrust Bearings: Designed to handle forces **parallel** to the shaft. The entire weight of the aluminum/steel linkages, motors, and payload pushes straight down due to gravity. The thrust bearings sit beneath the rotational joints to **absorb heavy downward axial load**, providing a frictionless rolling surface that prevents the arm from grinding against the stationary base.
- B) Radial Bearings: Designed to handle forces **perpendicular** to the shaft. As the SCARA arm extends, the weight creates a severe bending moment that tries to pull the rotating shaft sideways. The radial bearings **absorb this lateral force**, maintaining strict shaft concentricity.
- C) Linear Bearings: Paired with the guide rods. These provide a high-precision, **low-friction interface** for vertical movement, ensuring the Z-axis carriage moves smoothly **without jamming**.

MECHANICAL DESIGN

After going through different possible configurations, this particular architecture was selected due to the following reasons,

- X-Y planar movement is decoupled from the vertical Z-axis to minimize vertical deflection.
- Heavier NEMA-17 motors are closer to primary base pivot to reduce the load.
- The rotational joints have been isolated for control.
- The design, beyond pure kinematics, also has a clean visual aesthetic.

SHOP UTILIZATION & PROCESS JUSTIFICATION

Each shop was utilized to solve specific mechanical, material, or electrical constraints.

1. **Digital Manufacturing Lab:** Microcontroller programming, sensor integration
 - a) We utilized multimeter to precisely tune the potentiometers on our A4988 stepper drivers, setting the exact current limits required by the NEMA-17 motors to prevent thermal overload while ensuring maximum holding torque.
 - b) We integrated the color sensor's feedback loop, effectively giving the robot the decision-making capability required for the sorting operation.
2. **3D Printing Lab:** Rapid prototyping of gripper, motor couplers, pulleys and carriage
 - a) Machining the Z-axis carriage from a solid block of aluminum to include precise spaces for linear bearings, lead-nuts, and arm mounts would require a 5-axis CNC mill. 3D printing allowed us to consolidate these complex mounting features into a single, highly accurate part.
 - b) Printing the pulleys allowed us to achieve the exact tooth-profiles and custom sizes in accordance with our timing belt.
 - c) 3D printing the motor-to-shaft couplers provided material flexibility compared to solid steel, acting as a buffer to absorb minor angular misalignments.
 - d) Utilizing lightweight thermoplastics for the end-effector drastically minimized the mass at the furthest point of the arm, reducing load on the base motors.
3. **Machine Shop:** Used for precise diameter reduction of aluminum rods in support structure
 - a) Strict concentricity is required to prevent binding during rotation.
 - b) Single point turning tool was used to achieve exact tolerance & smooth surface finish.
4. **Fabrication Shop:** Used for rod cutting, drilling and mechanical assembly
 - a) We utilized a power hacksaw rather than manual cutting to cut the thick stainless-steel stock. The machine's rigid blade path provided a square, perpendicular rough cut.
 - b) A stationary bench drilling machine was mandatory for creating the linkage holes. Unlike hand-drilling, the machine's rigid column ensures absolute vertical perpendicularity.
 - c) We chose bolted mechanical fasteners over permanent welding for the main frame assembly. This allowed us to disassemble and adjust the linkages. Welding introduces extreme localized heat, causing thermal distortion and warping. However, if the main Z-axis pillar warps even one degree off-center, the entire planar geometry is ruined.
5. **Woodworking Shop:** Fabricating the heavy core of the base plank and constructing the Arduino enclosure.
 - a) Wood's high mass and vibration-dampening characteristics make it an ideal core for the base to anchor the shifting center-of-gravity.
 - b) Its natural dielectric properties allowed us to safely house the high-current stepper drivers and microcontroller without the risk of accidental short circuits inherent to metal casings.
6. **Sheet Metal Shop:** Cutting & shaping the metallic top-plate for the robot's base
 - a) Wood is compressible. Bolting the heavy steel Z-axis pillar directly to a purely wooden base would result in the bolts crushing the fibers over time, causing the main pillar to tilt.
 - b) We utilized sheet metal to clad the base, creating a rigid, non-compressible mounting interface that evenly distributes the localized stress of the heavy bolts.

MANUFACTURING INNOVATION

The primary innovation of this project was elevating the SCARA from a blind manipulator into an automated, decision-making sorting system. By integrating a TCS34725 RGB Color Sensor, we changed the fundamental control architecture of our prototype from an open-loop execution to a closed-loop sorting algorithm.

- Standard SCARA prototypes operate entirely open-loop, repeatedly moving to pre-programmed X-Y coordinates.
- By integrating the color sensor, we introduced an optical feedback loop. The microcontroller now reads environmental data and dynamically alters the robot's path based on algorithm, allowing it to actively categorize and physically sort objects by color.
- The TCS34725 was deliberately chosen over basic analog light sensors due to its integrated IR-blocking filter and onboard white LED. The IR filter ensures accurate color readings regardless of the room's lighting conditions.

Utility: In continuous manufacturing, products often come down a high-speed conveyor. A vision-integrated SCARA acts as an automated quality inspector. It can detect parts that have been painted the wrong color, or mislabeled products, and immediately eject them from the assembly line without human intervention. Alternatively, it can be used in high-speed logistics to read labels, sorting harvested products based on ripeness and pharmaceutical packaging for distinctly colored pills.

OPERATIONAL DYNAMICS

1. Kinematics Control Systems: Working principle of control systems
 - a) Forward Kinematics (State Feedback): This is the process of calculating the exact position of the gripper (X, Y, Z) based on the known, current rotational angles of our stepper motors. It helps us identify where gripper is located in physical space.
 - b) Inverse Kinematics (Path feedback): When the color sensor identifies a target, it generates a desired (X, Y) spatial coordinate. Inverse kinematics works backward, using trigonometric principles and the known lengths of our arms to calculate the exact joint angles required to reach that target.
2. Selective Compliance: Working principle of mechanical design
 - a) Rigid in the z-axis: When lead screw pushes the carriage down, the support rods ensure the arms do not flex upward, allowing it to apply strong downward insertion force without deflecting.
 - b) Compliant in the x-y plane: The horizontal linkages are designed to swing freely.